PackRisk

Battery-pack supply risk: where a mitigation budget protects the most production

What it is

- A simulated pack line (50 parts, 20 suppliers, 4 shipping lanes) faces 11 disruption scenarios, each run 100 times.
- Seven strategies spend a $250K budget at week 0 on expediting, extra stock or approved alternate suppliers.
- Built in Python (stochastic optimization), SQL and Power BI. All operational data is simulated.

Key results

- Asia port congestion: lost packs 1,037 → 75; about $2.4M net saved for $181K of spend (13.1x).
- Beat doing nothing on all 5 held-out test scenarios, and in 33 of 33 cases with a lost pack valued at $1K-$4K.
- Late rate alone is a poor guide: among the latest suppliers, only parts with 2 weeks of stock stopped the line.
- Rules of thumb can backfire: protecting the lowest-stock parts first cost more than doing nothing.

In this PDF

- Pages 2-12: Executive Overview for each of the 11 scenarios (D1-D6 development, H1-H5 unseen tests).
- Pages 13-16: Supplier Performance, Production Risk, Scenario Explorer, Assumptions (Asia port congestion).

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

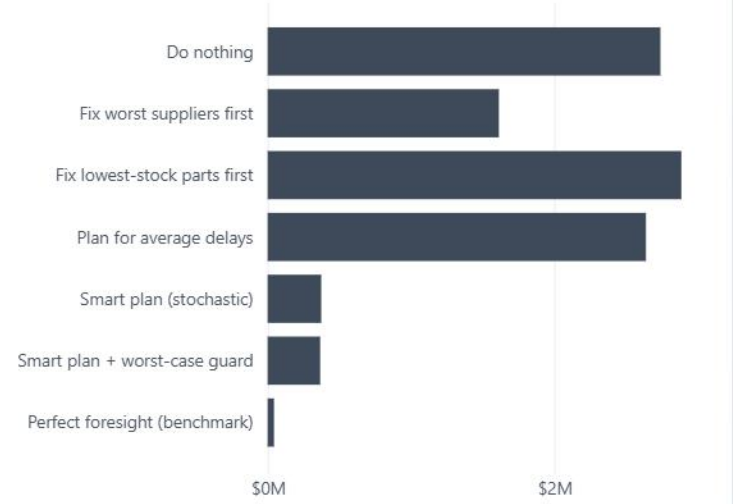
Scenario: Asia ocean-lane port congestion

The smart plan protects 963 packs and saves $2.4M net for $181K of fixes (13.1x return).

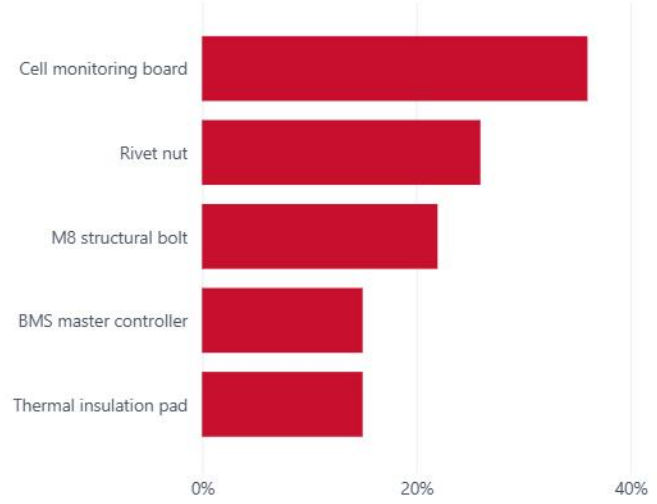
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
1,037	75	963	$2.4M	$181.2K	13.1x	93.1%

All L1 ocean shipments delayed; hits both cell suppliers, electronics, contactors, fire barriers, fasteners.

Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Development scenario

Bad case (worst 10%): do nothing
2,600

Bad case (worst 10%): smart plan
106

Simulated data, placeholder delay parameters

Scenario D1: Asia ocean-lane port congestion (development)

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

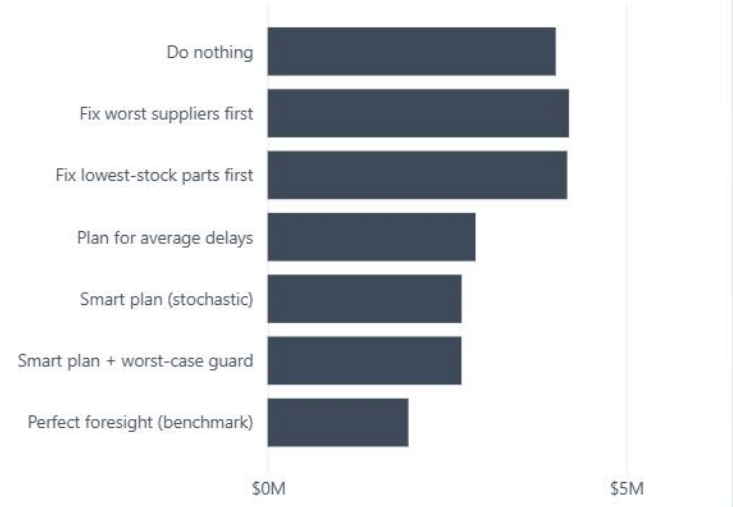
Scenario: Semiconductor allocation

The smart plan protects 702 packs and saves $1.3M net for $250K of fixes (5.3x return).

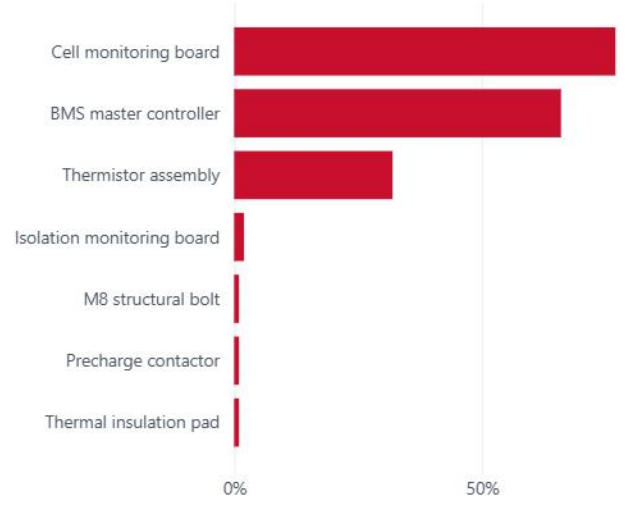
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
1,738	1,035	702	$1.3M	$250.0K	5.3x	76.5%

Tier-2 semiconductor shortage cuts output at the electronics and sensor suppliers together.

Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Development scenario

Bad case (worst 10%): do nothing
3,474

Bad case (worst 10%): smart plan
2,549

Simulated data. Placeholder delay parameters.

Scenario D2: Semiconductor allocation (development)

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

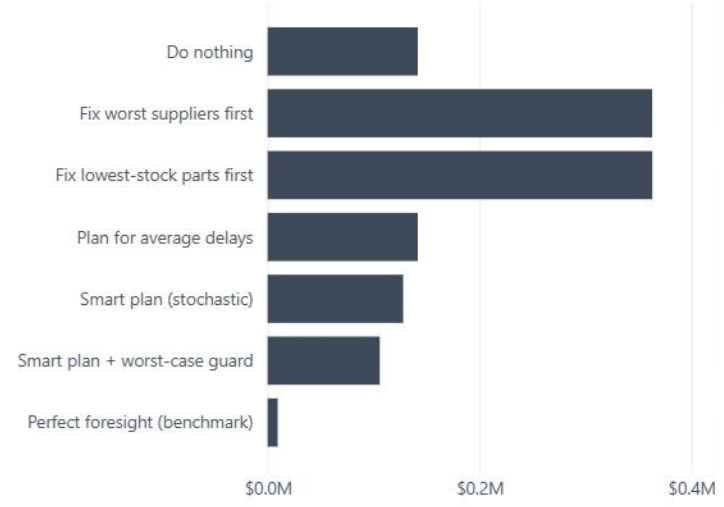
Scenario: Enclosure plant outage

The smart plan protects 14 packs and saves $14K net for $18K of fixes (0.8x return).

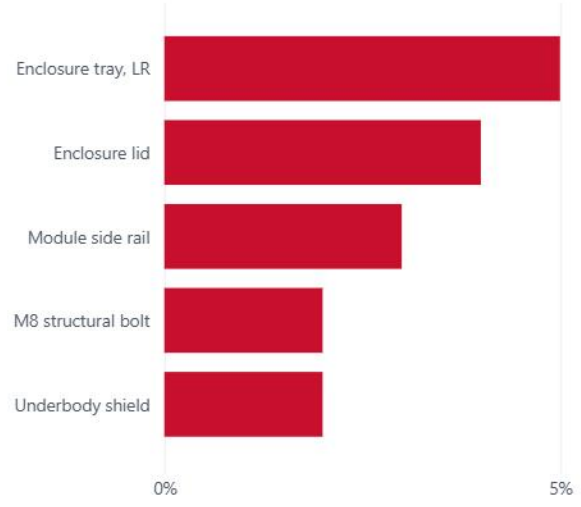
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
60	46	14	$13.8K	$18.0K	0.8x	25.3%

Single-supplier capacity loss at the enclosure supplier.

Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Development scenario

Bad case (worst 10%): do nothing
4

Bad case (worst 10%): smart plan
0

Simulated data. Placeholder delay parameters.

Scenario D3: Enclosure plant outage (development)

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

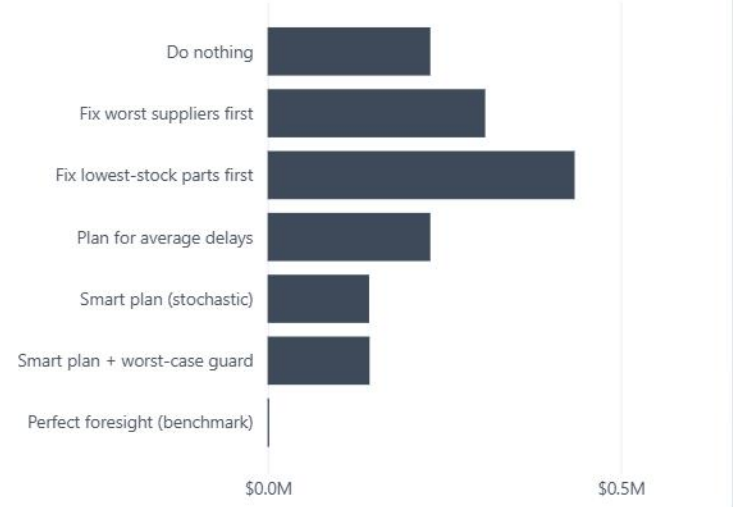
Scenario: TIM quality hold

The smart plan protects 38 packs and saves $87K net for $3K of fixes (26.4x return).

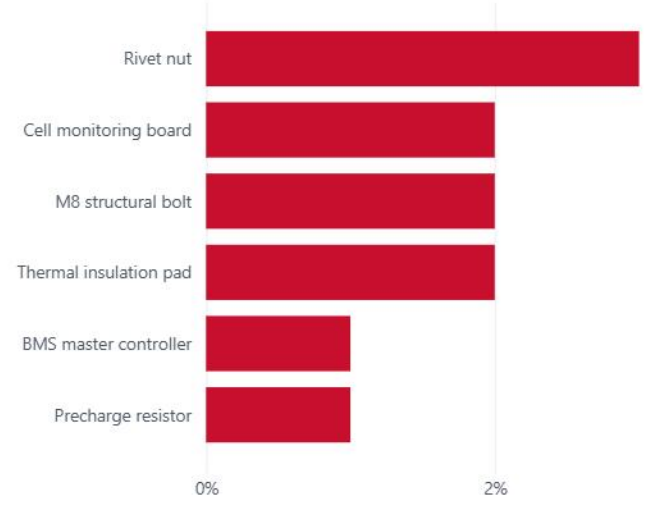
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
97	58	38	$86.7K	$3,278	26.4x	39.7%

Thermal interface material lots quarantined, part scrapped.

Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Development scenario

Bad case (worst 10%): do nothing
0

Bad case (worst 10%): smart plan
0

Simulated data. Placeholder delay parameters.

Scenario D4: TIM quality hold (development)

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

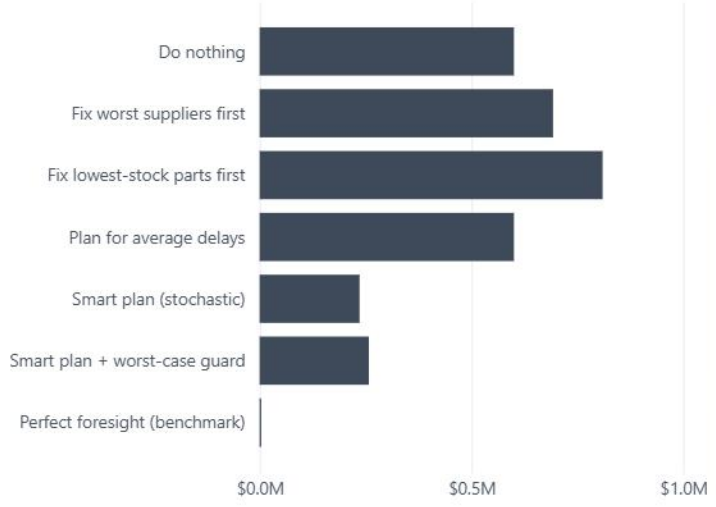
Scenario: Demand mix shift

The smart plan protects 172 packs and saves $365K net for $34K of fixes (10.8x return).

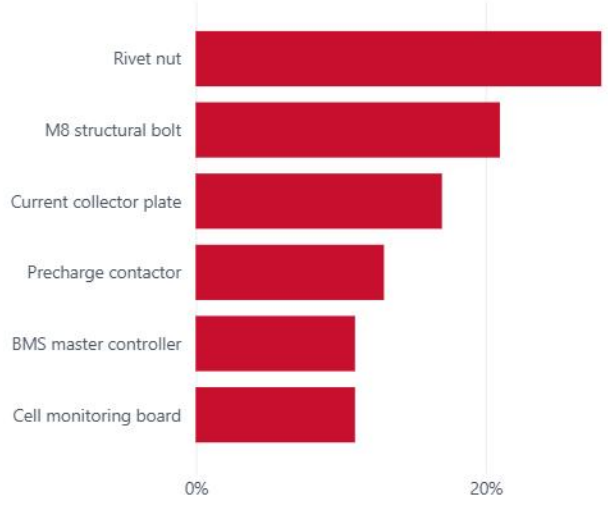
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
257	84	172	$364.5K	$33.8K	10.8x	67.2%

Long Range plan +20%, Standard Range -10%; purchase orders still follow the old plan.

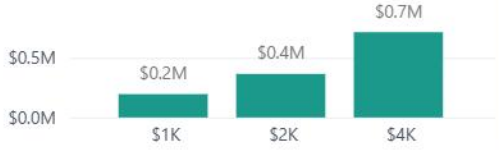
Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Development scenario

Bad case (worst 10%): do nothing
540

Bad case (worst 10%): smart plan
141

Simulated data. Placeholder delay parameters.

Scenario D5: Demand mix shift (development)

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

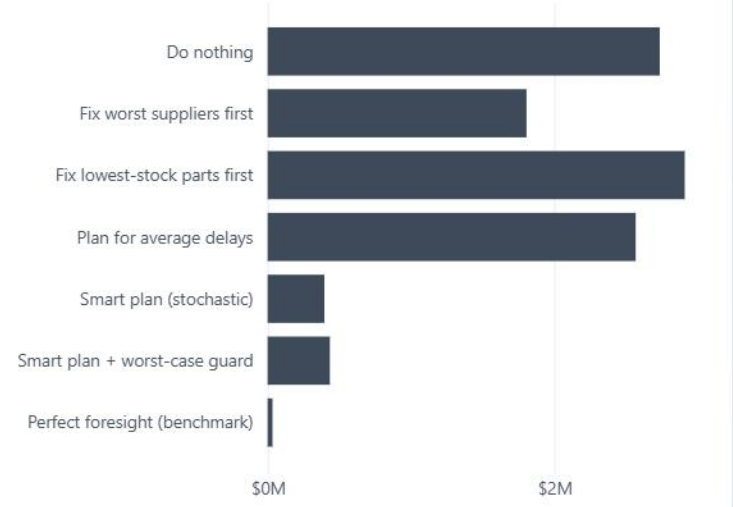
Scenario: Port congestion + TIM hold

The smart plan protects 938 packs and saves $2.3M net for $174K of fixes (13.5x return).

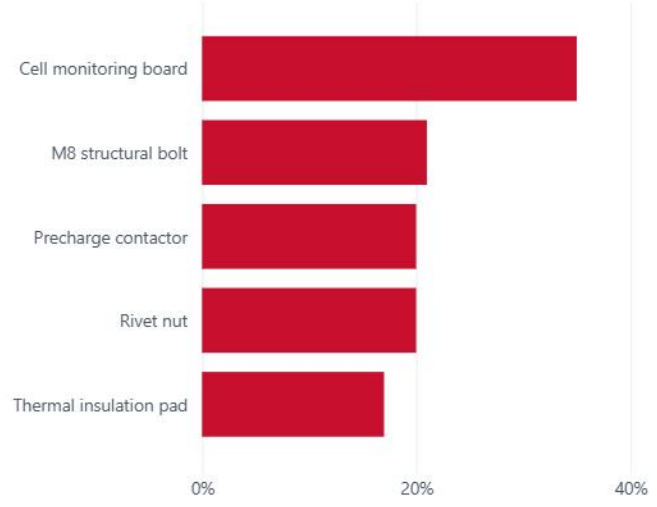
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
1,032	93	938	$2.3M	$173.8K	13.5x	90.9%

D1 and D4 at the same time.

Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Development scenario

Bad case (worst 10%): do nothing
2,600

Bad case (worst 10%): smart plan
28

Simulated data. Placeholder delay parameters.

Scenario D6: Port congestion + TIM hold (development)

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

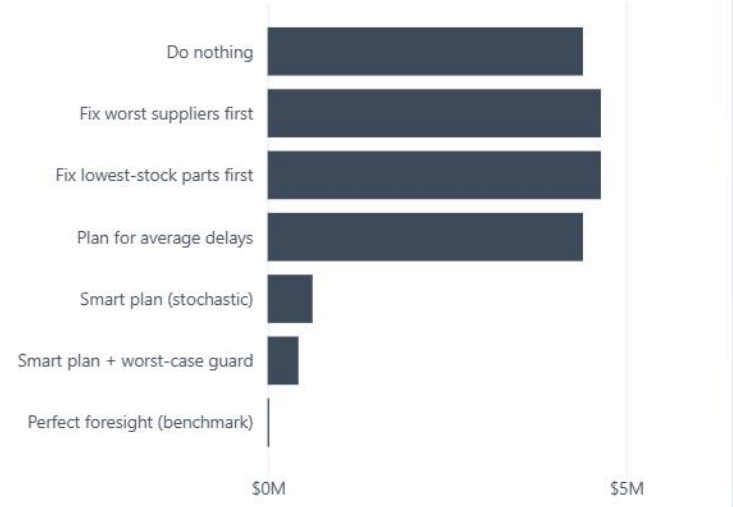
Scenario: Europe ocean-lane disruption

The smart plan protects 1,456 packs and saves $3.8M net for $39K of fixes (96.6x return).

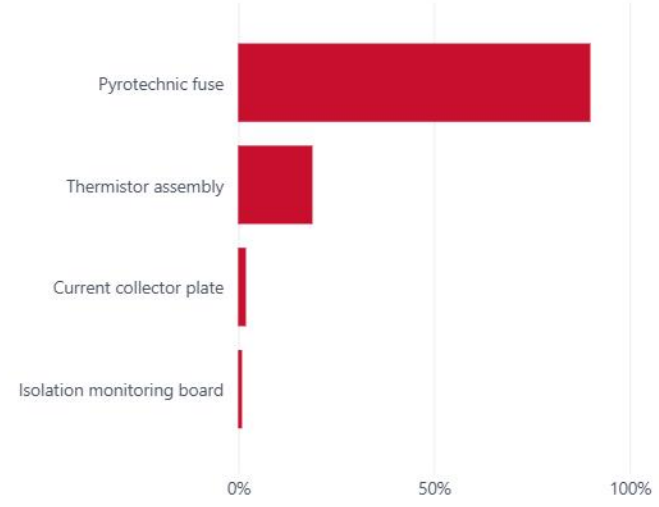
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
1,674	218	1,456	$3.8M	$39.1K	96.6x	87.0%

L4 delays longer and more severe than the planner's prior.

Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Unseen test (held out)

Bad case (worst 10%): do nothing
2,610

Bad case (worst 10%): smart plan
580

Simulated data. Placeholder delay parameters.

Scenario H1: Europe ocean-lane disruption (unseen test)

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

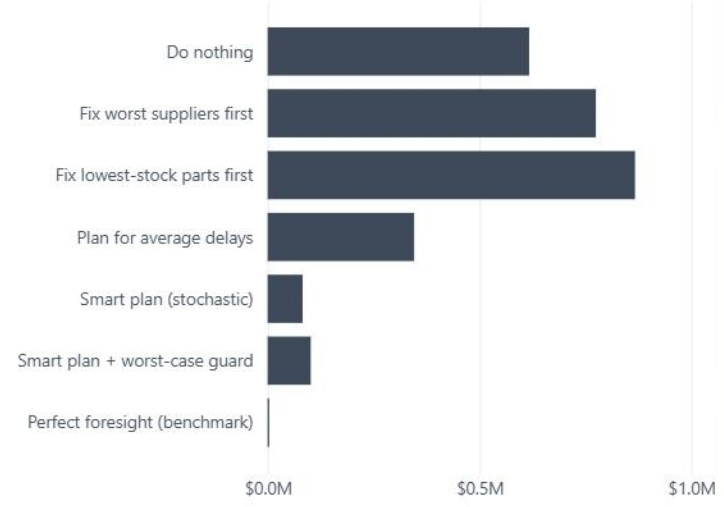
Scenario: Copper shortage

The smart plan protects 240 packs and saves $535K net for $35K of fixes (15.3x return).

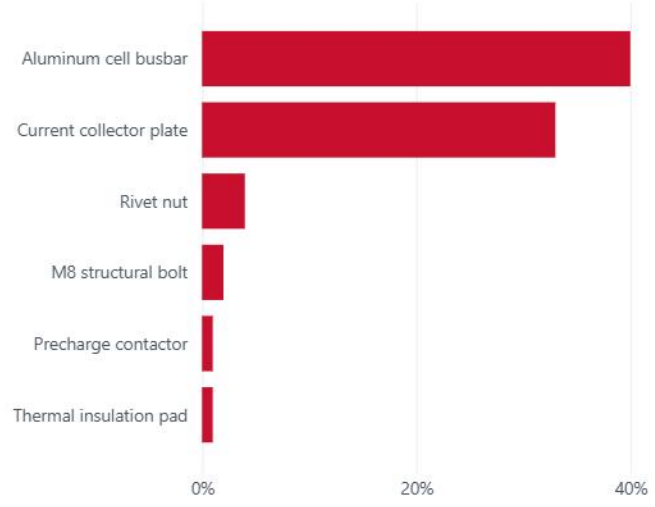
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
258	19	240	$534.8K	$35.0K	15.3x	92.7%

Tier-2 copper cut hits busbar and harness suppliers together; shorter than prior expects.

Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Unseen test (held out)

Bad case (worst 10%): do nothing
724

Bad case (worst 10%): smart plan
0

Simulated data. Placeholder delay parameters.

Scenario H2: Copper shortage (unseen test)

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

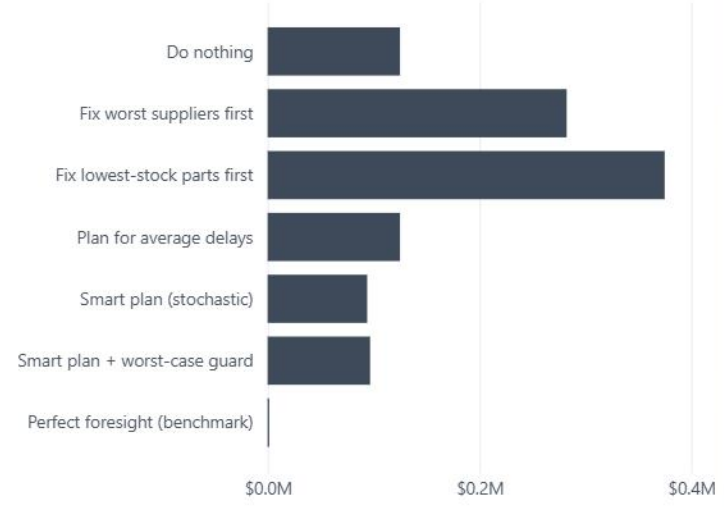
Scenario: Long adhesive quality hold

The smart plan protects 15 packs and saves $31K net for $6K of fixes (5.2x return).

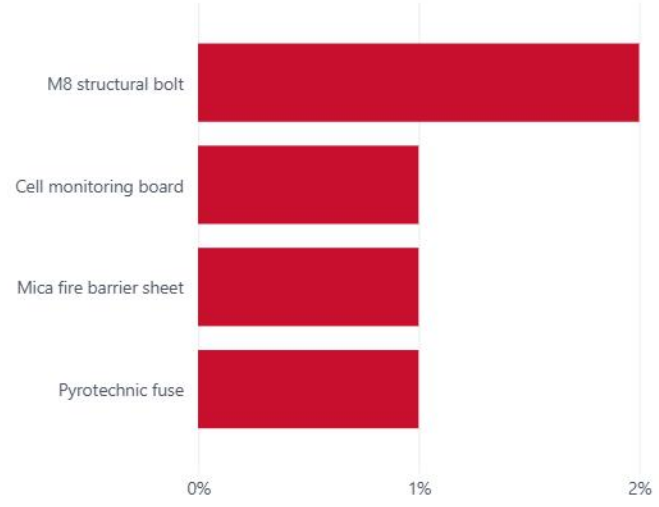
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
52	37	15	$31.0K	$5,913	5.2x	28.7%

4-week hold with 50% scrap on structural adhesive, outside the planner's prior.

Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Unseen test (held out)

Bad case (worst 10%): do nothing
0

Bad case (worst 10%): smart plan
0

Simulated data. Placeholder delay parameters.

Scenario H3: Long adhesive quality hold (unseen test)

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

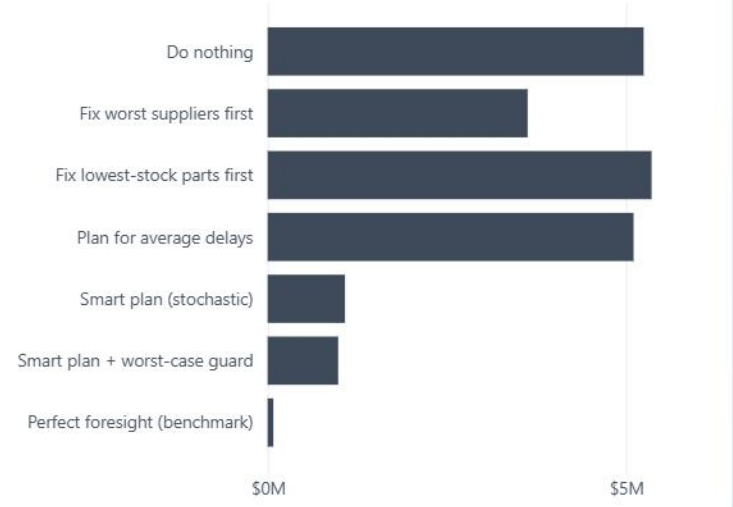
Scenario: Port congestion, 30% worse

The smart plan protects 1,655 packs and saves $4.2M net for $187K of fixes (22.3x return).

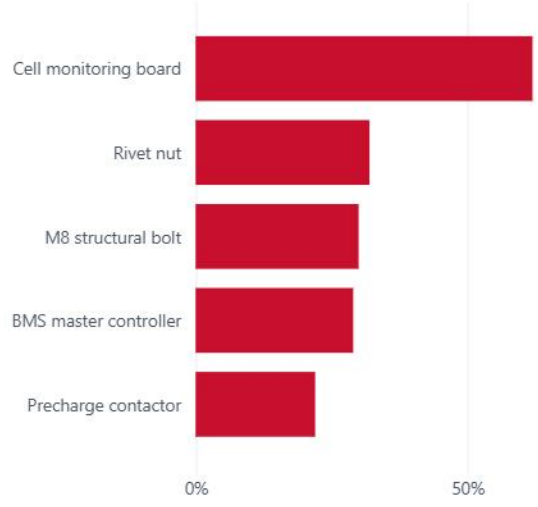
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
2,024	368	1,655	$4.2M	$187.0K	22.3x	81.8%

D1, but event delays and background lateness are 30% worse than the planner assumes.

Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Unseen test (held out)

Bad case (worst 10%): do nothing
2,981

Bad case (worst 10%): smart plan
1,049

Simulated data. Placeholder delay parameters.

Scenario H4: Port congestion, 30% worse (unseen test)

PACKRISK
SUPPLY RISK ANALYTICS

Executive Overview

Smart plan vs doing nothing for the selected scenario. On all 5 unseen test scenarios the smart plan won 5 of 5 (716 packs saved on average, $9.1M net).

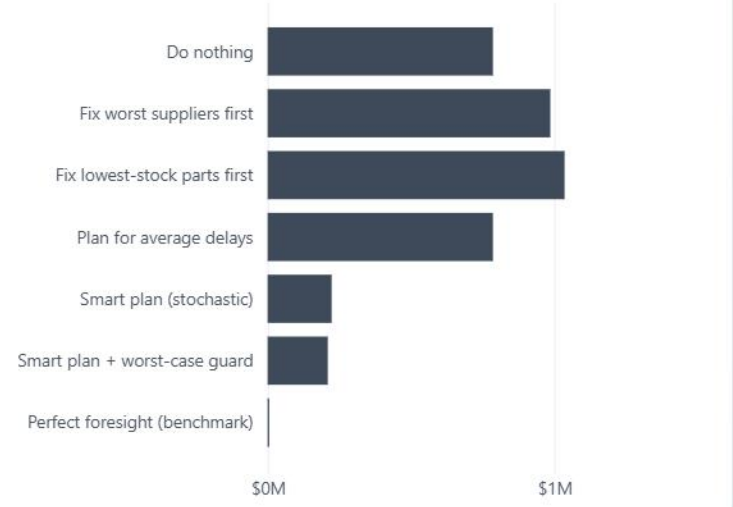
Scenario: Multi-event combination

The smart plan protects 212 packs and saves $563K net for $22K of fixes (25.8x return).

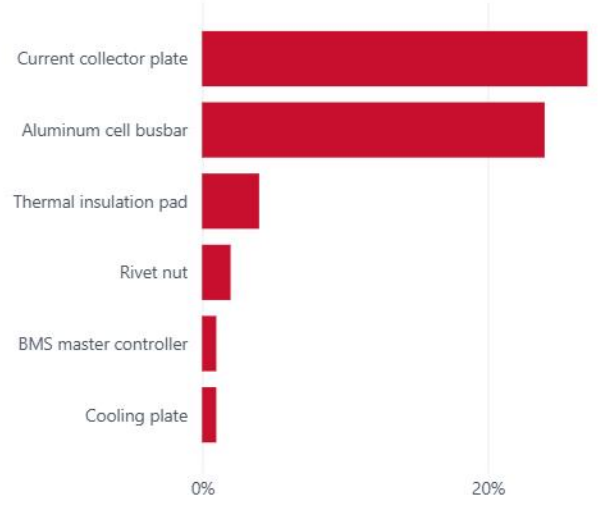
Lost: do nothing	Lost: smart plan	Packs protected	Net savings	Spend on fixes	Saved per $1 spent	Share of best possible
295	83	212	$562.8K	$21.8K	25.8x	71.8%

Mexico truck-lane delay + connector supplier outage + potting foam hold, together.

Expected total cost by strategy (lower is better)



Top 5 parts that stop the line (if we do nothing)



Net savings by value of a lost pack



This scenario is a
Unseen test (held out)

Bad case (worst 10%): do nothing
1,100

Bad case (worst 10%): smart plan
65

Simulated data. Placeholder delay parameters.

Scenario H5: Multi-event combination (unseen test)

PACKRISK
SUPPLY RISK ANALYTICS

Supplier Performance

Supplier history is fixed; everything on this page shows its impact in the selected scenario (if we do nothing).

Scenario: Asia ocean-lane port congestion

Parts that stop the line	Suppliers whose parts stop it	Their avg late rate (all: 28.8%)	Worst part stops the line
8	4	54.2%	36%

Supplier scorecard: history + how often its parts stop the line here

Supplier	Supplies	Lane	OTIF	Late rate	Lead-time SD	Stops line here
S07	Electronics (CMB, BMS master)	L1	34.6%	57.7%	6.49	36%
S18	Fasteners	L1	35.9%	57.7%	6.13	26%
S15	Mica, aerogel barriers	L1	41.0%	57.7%	5.76	15%
S09	Contactors, precharge parts	L1	52.6%	43.6%	6.35	12%
S01	LFP prismatic cells	L1	50.0%	46.2%	4.77	0%
S02	LFP prismatic cells (alternate)	L1				0%
S03	NMC cylindrical cells	L1	42.3%	57.7%	6.25	0%
S04	NMC cylindrical cells (alternate)	L3				0%
S05	Busbars, current collectors	L2	74.0%	17.3%	3.55	0%
S06	HV cables, harnesses	L2	61.5%	29.5%	3.40	0%

Late rate vs lead-time SD (bubble = stops line here)



Shared lanes and tier-2 sources: which one hurts here

Type	ID	Suppliers	Parts	Single-src	Weekly spend	Stops line here
Shipping lane	L1	6	13	6	$10,690,200	36%
Tier-2 source	T2	2	5	5	$645,600	36%
Tier-2 source	T5	1	3	3	$196,200	15%
Shipping lane	L2	4	11	11	$733,200	0%
Shipping lane	L3	6	21	19	$1,163,490	0%
Shipping lane	L4	2	5	5	$233,100	0%
Tier-2 source	T1	2	7	7	$390,300	0%

KEY INSIGHT: parts that stop the line in this scenario (do nothing)

Supplier	Part	Stops line	Late rate	Stock (weeks)
S07	Cell monitoring board	36%	57.7%	2.00
S18	Rivet nut	26%	57.7%	2.00
S18	M8 structural bolt	22%	57.7%	2.00
S07	BMS master controller	15%	57.7%	2.00
S15	Thermal insulation pad	15%	57.7%	2.00
S09	Precharge contactor	12%	43.6%	2.00
S15	Mica fire barrier sheet	2%	57.7%	3.00

Simulated data. Placeholder delay parameters.

PACKRISK
SUPPLY RISK ANALYTICS

Production Risk

Severe (P90) realization is the default: in the median realization most policies show no shortfall.

Scenario: Asia ocean-lane port congestion

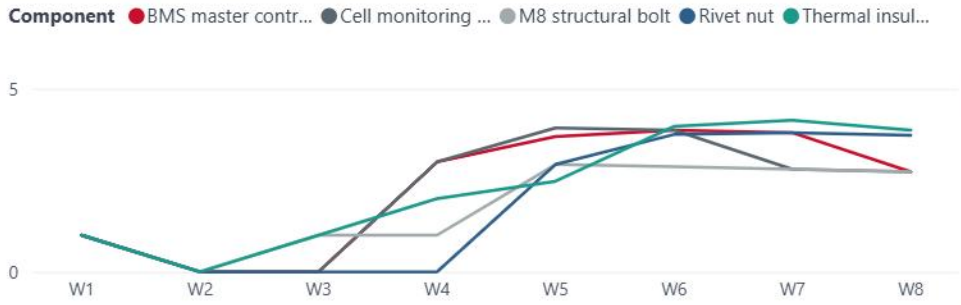
Outcome: Bad case (worst 10%)

Share of plan built (do nothing)	Lost packs, bad case (do nothing)	Backlog pack-weeks (do nothing)
91.4%	2,600	6,670

Cumulative shortfall vs plan (packs)



Weeks of stock left: top 5 problem parts (do nothing)



How often each part stops the line, by week (do nothing; SR + LR added together)

Component	W2	W3	W4	W5	W6	W7	W8	Total
BMS master controller		11%	13%	3%				**27%**
Cell monitoring board	5%	52%	22%	4%	2%	2%		**87%**
M8 structural bolt		23%	10%	8%	3%	2%	2%	**48%**
Main HV contactor			1%					**1%**
Mica fire barrier sheet			3%					**3%**
Precharge contactor		13%	6%	2%	2%			**23%**
Rivet nut		11%	21%	13%	3%	1%	1%	**50%**
Thermal insulation pad		15%	9%	1%				**25%**
Total	**5%**	**125%**	**85%**	**31%**	**10%**	**5%**	**3%**	**264%**

Simulated data. Placeholder delay parameters.

PACKRISK
SUPPLY RISK ANALYTICS

Scenario Explorer

Pick strategies to compare and see exactly what each plan buys in the selected scenario.

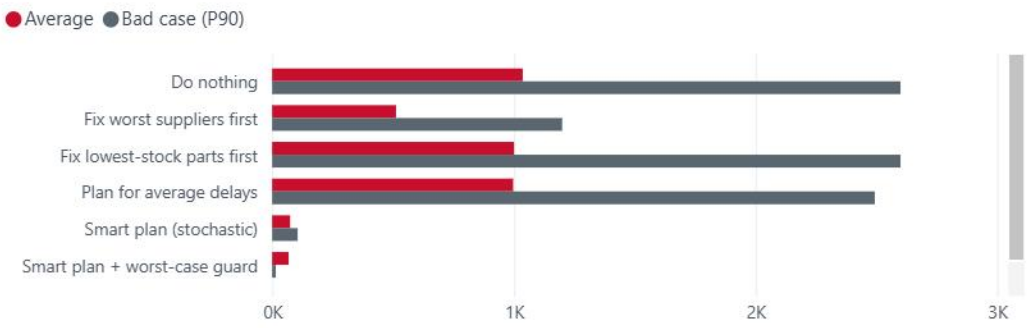
Scenario

Asia ocean-lane port congestion

Strategy (pick any)

- [ ] Do nothing
- [ ] Fix worst suppliers first
- [ ] Fix lowest-stock parts first
- [ ] Plan for average delays
- [ ] Smart plan (stochastic)
- [ ] Smart plan + worst-case guard
- [ ] Perfect foresight (benchmark)

Lost packs by strategy: average vs bad case (worst 10%)



Value of planning for uncertainty

$2.3M

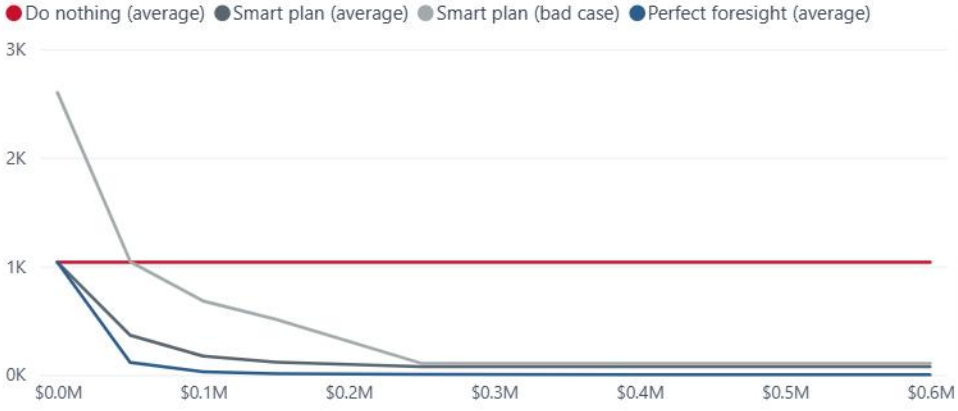
Budget needed (more barely helps)

About $250K

What the plan actually buys (empty = buys nothing in this scenario)

Strategy	ID	Component	Action	Week	Quantity	Cost
Fix worst suppliers first	C11	Cell monitoring board	expedite	1	6,000	$40,500
Fix worst suppliers first	C30	Thermal insulation pad	expedite	1	12,600	$3,780
Fix worst suppliers first	C31	Mica fire barrier sheet	expedite	1	6,000	$5,400
Fix worst suppliers first	C32	Aerogel cell spacer	expedite	1	45,000	$20,250
Fix worst suppliers first	C42	M6 flange bolt	expedite	1	90,000	$1,350
Fix worst suppliers first	C43	M8 structural bolt	expedite	1	36,000	$1,350
Fix worst suppliers first	C44	Rivet nut	expedite	1	60,000	$1,350
Fix lowest-stock parts first	C11	Cell monitoring board	expedite	1	6,000	$40,500
Fix lowest-stock parts first	C12	BMS master controller	expedite	1	1,500	$36,000
Smart plan (stochastic)	C11	Cell monitoring board	expedite	1	5,200	$35,100
Smart plan (stochastic)	C14	Precharge contactor	expedite	1	912	$3,009
Smart plan (stochastic)	C15	Precharge resistor	expedite	1	5	$4
Smart plan (stochastic)	C43	M8 structural bolt	expedite	1	28,686	$1,076

Lost packs vs mitigation budget (fixed strategies; not affected by the picker)



Simulated data. Placeholder delay parameters.

PACKRISK
SUPPLY RISK ANALYTICS

Assumptions & Limitations

The assumptions apply to every scenario. The two tables at the bottom change with the scenario.

Scenario: Asia ocean-lane port congestion

Static week-0 decisions (no rolling re-planning). Delay parameters are PLACEHOLDERS, not calibrated. Costs and penalties ($2,000 per lost pack, $100 per backlog pack-week, $250,000 budget) are hypothetical.

1. Site, products, plan

- One hypothetical site, weekly buckets, 8-week horizon; SR (LFP) and LR (NMC) packs.
- Plan SR 900 + LR 600 per week (12,000 packs); capacity 1,600/week.
- Lost packs = plan - built at week 8; backlog pack-weeks measure how far and how long production runs behind.

2. Supply

- 20 tier-1 suppliers, 6 tier-2 sources, 4 lanes, 50 components; just-in-time weekly POs.
- Lots can be late, partial, quality-held, or cut by capacity loss / scrap; arrivals after week 8 ignored.

3. Disruptions

- Planner knows which disruption was announced, not its severity or duration.
- D1-D6: truth equals prior. H1-H5: held out, truth deliberately differs from prior.

4. Mitigation actions

- Expedite (air or team driver), extra inventory, approved alternate supplier.
- Only incremental mitigation cost counts against the $250,000 budget.

5. Decisions and evaluation

- Static two-stage: all actions decided once at week 0.
- Deterministic LP, Stochastic LP (100 samples), CVaR variant; oracle solves the same LP with perfect knowledge.
- Planning (seed 42) and evaluation (seed 2027) samples are independent.
- Penalty test: at $1K, $2K and $4K per lost pack, the smart plan beats doing nothing in 33 of 33 cases.

6. Known limitations

- Simulated operations; placeholder delays; calibration source is not battery parts.
- No rolling re-planning, no backlog recovery after capacity cuts, no MOQs or integer lots.

Does modelling shared shocks help here? (lost packs)



Correlation stress tests for this scenario (all numbers)

test	scenario_id	setting	metric	value
Correlation (do nothing)	D1	correlated	mean_lost_packs	1,033.63
Correlation (do nothing)	D1	correlated	p90_lost_packs	2,600.00
Correlation (do nothing)	D1	correlated	p95_lost_packs	2,700.00
Correlation (do nothing)	D1	correlated	prob_any_loss	0.73
Correlation (do nothing)	D1	independent	mean_lost_packs	1,294.56
Correlation (do nothing)	D1	independent	p90_lost_packs	2,600.00
Correlation (do nothing)	D1	independent	p95_lost_packs	2,700.00
Correlation (do nothing)	D1	independent	prob_any_loss	0.86
Planner correlation assumption	D1	planner assumes correlation	mean_lost_packs	74.52

What the smart plan buys if it assumes shared vs indep...

Policy	Action	Quantity	Cost
P5-corr	alternate	64,800	$864
P5-corr	expedite	231,816	$180,382
P5-indep	alternate	28,800	$432
P5-indep	expedite	309,080	$174,573
P5-indep	extra	2,820	$784